Math 20

From UPD IM Module, 2018 — Module 0

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0 Notes

0.1 Fractions

Note 0.1.1. Addition and Subtraction. Let L be the LCD of the fractions.

$$\frac{a}{b} \pm \frac{c}{d} = \frac{a\left(\frac{L}{b}\right) \pm c\left(\frac{L}{d}\right)}{L}$$

Note 0.1.2. Multiplication.

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

Note 0.1.3. Division.

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} = \frac{ad}{bc}$$

If both fractions have the same denominator:

$$\frac{a}{b} \div \frac{c}{b} = \frac{a}{b} \cdot \frac{b}{c} = \frac{ab}{bc} = \frac{a}{c}$$

0.2 Exponents

Note 0.2.1. Negative Exponents.

$$a^{-n} = \frac{1}{a^n}$$

Note 0.2.2. Multiplication.

$$a^n a^m = a^{n+m}$$

Note 0.2.3. Division.

$$\frac{a^n}{a^m} = a^{n-m}$$

Note 0.2.4. Of Products.

$$(ab)^n = a^n b^n$$

Note 0.2.5. Of Quotients.

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

Note 0.2.6. Nesting.

$$(a^n)^m = a^{nm}$$

0.3 Factoring

Note 0.3.1. Common Monomial Factor.

$$ax \pm ay = a(x \pm y)$$

Note 0.3.2. Difference of Two Squares.

$$x^2 - y^2 = (x - y)(x + y)$$

Note 0.3.3. Perfect Square Trinomial.

$$x^2 \pm 2xy + y^2 = (x \pm y)^2$$

Note 0.3.4. Sum and Difference of Two Cubes.

$$x^3 \pm y^3 = (x \pm y)(x^2 \mp xy + y^2)$$

Note 0.3.5. Simple Trinomial.

$$x^2 + (a + b)x + ab = (x + a)(x + b)$$

Note 0.3.6. Complex Trinomial.

$$acx^2 + (ad + bc)x + bd = (ax + b)(cx + d)$$

0.4 Radicals

Note 0.4.1. Exponent Form.

$$\sqrt[n]{a^m} = a^{\frac{m}{n}}$$

Note 0.4.2. Of Products.

$$\sqrt[n]{ab} = \sqrt[n]{a}\,\sqrt[n]{b}$$

Note 0.4.3. Of Quotients.

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

Note 0.4.4. Nesting.

$$\sqrt[n]{\sqrt[m]{a}} = \sqrt[nm]{a}$$

Note 0.4.5. Rationalization. A rational expression must not have a radical at its denominator. Multiply the numerator and denominator by an expression that would eliminate the radical; this may be a special product. A simple example:

$$\begin{aligned}
\sqrt[3]{\frac{10y^4}{3x^2}} &= \frac{\sqrt[3]{10y^4}}{\sqrt[3]{3x^2}} \\
&= \frac{\sqrt[3]{10y^4}}{\sqrt[3]{3x^2}} \cdot \frac{\sqrt[3]{9x}}{\sqrt[3]{3^2x}} \\
&= \frac{\sqrt[3]{90xy^4}}{\sqrt[3]{3^3x^3}} \\
&= \frac{\sqrt[3]{y^3}\,\sqrt[3]{90xy}}{3x} \\
&\boxed{= \frac{y\sqrt[3]{90xy}}{3x}}
\end{aligned}$$

A more complex example using note 0.3.4:

$$\begin{aligned}
\frac{1}{2-\sqrt[3]{4}} &= \frac{1}{2-\sqrt[3]{4}} \cdot \frac{4+2\sqrt[3]{4}+\sqrt[3]{16}}{4+2\sqrt[3]{4}+\sqrt[3]{4^2}} \\
&= \frac{4+2\sqrt[3]{4}+\sqrt[3]{8}\,\sqrt[3]{2}}{2^3-\sqrt[3]{4^3}} \\
&= \frac{4+2\sqrt[3]{4}+2\sqrt[3]{2}}{8-4} \\
&= \frac{4+2\sqrt[3]{4}+2\sqrt[3]{2}}{4} \\
&\boxed{= 1+\frac{\sqrt[3]{4}+\sqrt[3]{2}}{2}}
\end{aligned}$$

0.5 Imaginary Numbers

Note 0.5.1. Definition.

$$i = \sqrt{-1}$$

Note 0.5.2. Cyclicity. We have:

$$\begin{aligned} i^0 &= 1 \\ i^1 &= i \\ i^2 &= -1 \\ i^3 &= -i \end{aligned}$$

This is cyclic:

$$i^n = i^{n \bmod 4}$$

0.6 Complex Numbers

Note 0.6.1. Definition. A complex number in rectangular form is given by:

$$a + bi$$

Note 0.6.2. Addition.

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

Note 0.6.3. Multiplication.

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

Note 0.6.4. Conjucate.

$$\overline{a + bi} = a - bi$$

Also,

$$(a + bi)(\overline{a + bi}) = a^2 + b^2$$

Note 0.6.5. Division.

$$\frac{a + bi}{c + di} = \frac{a + bi}{c + di} \cdot \frac{\overline{c + di}}{\overline{c + di}} = \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2} i$$

1

Simplify the following.

1.1

$$\frac{7^{-3}}{7^{-16}}$$

Note 0.2.3

$$= 7^{-3+16}$$

$$\boxed{= 7^{13}}$$

1.2

$$\frac{3^2 + 9^{-1} - 9^0}{2^4 - 3^{-1}}$$

$$= \left(9 + \frac{1}{9} - 1\right) \div \left(16 - \frac{1}{3}\right)$$

Notes 0.1.1 and 0.1.3

$$= \frac{81 + 1 - 9}{9} \cdot \frac{3}{48 - 1}$$
$$= \frac{73}{3} \cdot \frac{1}{47}$$

Note 0.1.2

$$\boxed{= \frac{73}{141}}$$

1.3

$$\frac{8x^6y^3z^4}{2x^5yz^2}$$

Note 0.2.3

$$= 4x^{6-5}y^{3-1}z^{4-2}$$

$$\boxed{= 4xy^2z^2}$$

1.4

$$\frac{3x^{21}y^{12}z^{19}}{(-x^{-9}y^{3}z^{-3})(-9x^{4}y^{6}z^{6})}$$

Note 0.2.2

$$= \frac{3x^{21}y^{12}z^{19}}{9x^{-9+4}y^{3+6}z^{-3+6}}$$
$$= \frac{3x^{21}y^{12}z^{19}}{9x^{-5}y^{9}z^{3}}$$

Note 0.2.3

$$= \frac{1}{3}x^{21+5}y^{12-9}z^{19-3}$$

$$\boxed{= \frac{1}{3}x^{26}y^{3}z^{16}}$$

1.5

$$\frac{2(x^3yz^2)^2}{(2x^2y^2z)^3}$$

Notes 0.2.4 and 0.2.6

$$= \frac{2x^6y^2z^4}{8x^6y^6z^3}$$

Note 0.2.3

$$= \frac{1}{4}x^{6-6}y^{2-6}z^{4-3}$$
$$= \frac{1}{4}y^{-4}z$$

Note 0.2.1

$$\boxed{= \frac{z}{4y^4}}$$

1.6

$$\frac{\left(2a^3b^{-3}c^{-2}\right)^3}{\left(a^3b^2c^{-1}\right)^{-2}}$$

Notes 0.2.4 and 0.2.6

$$= \frac{8a^9b^{-9}c^{-6}}{a^{-6}b^{-4}c^2}$$

Note 0.2.3

$$= 8a^{9+6}b^{-9+4}c^{-6-2}$$
$$= 8a^{15}b^{-5}c^{-8}$$

Note 0.2.1

$$\boxed{= \frac{8a^{15}}{b^5c^8}}$$

1.7

$$\left(\frac{2a^3b^{-3}c^{-1}}{3a^{-2}b^{-2}c^3}\right)^3$$

Note 0.2.3

$$= \left(\frac{2}{3}a^{3+2}b^{-3+2}c^{-1-3}\right)^3$$
$$= \left(\frac{2}{3}a^5b^{-1}c^{-4}\right)^3$$

Notes 0.2.4 to 0.2.6

$$= \frac{8}{27}a^{15}b^{-3}c^{-12}$$

Note 0.2.1

$$\boxed{= \frac{8a^{15}}{27b^3c^{12}}}$$

1.8

$$\left(\frac{a^2b^3c^{-1}}{ab^{-2}c^5}\right)^4$$

Note 0.2.3

$$= \left(a^{2-1}b^{3+2}c^{-1-5}\right)^4$$
$$= \left(ab^5c^{-6}\right)^4$$

Notes 0.2.4 and 0.2.6

$$= a^4b^{20}c^{-24}$$

Note 0.2.1

$$\boxed{= \frac{a^4b^{20}}{c^{24}}}$$

1.9

$$\left(\frac{4^0x^3y^{-2}z^{-4}}{3^2x^{-1}y^3z^{-5}}\right)^{-3}$$

$$= \left(\frac{x^3y^{-2}z^{-4}}{9x^{-1}y^3z^{-5}}\right)^{-3}$$

Note 0.2.3

$$= \left(\frac{1}{9}x^{3+1}y^{-2-3}z^{-4+5}\right)^{-3}$$

$$= \left(\frac{1}{9}x^4y^{-5}z\right)^{-3}$$

Notes 0.2.1 and 0.2.4 to 0.2.6

$$= 729x^{-12}y^{15}z^{-3}$$

Note 0.2.6

$$\boxed{= \frac{729y^{15}}{x^{12}z^3}}$$

1.10

$$\left(\frac{18^5 20^3 x^{-1} y^2 z^4}{15^9 x^{-3} y^{-2} z^{-1}}\right)^2$$

Notes 0.2.2 and 0.2.4

$$= \left(\frac{2^5 3^{10} 2^6 5^3 x^{-1} y^2 z^4}{3^9 5^9 x^{-3} y^{-2} z^{-1}}\right)^2$$

Note 0.2.2

$$= \left(\frac{2^{11} 3^{10} 5^3 x^{-1} y^2 z^4}{3^9 5^9 x^{-3} y^{-2} z^{-1}}\right)^2$$

Note 0.2.3

$$= (2^{11} 3^{10-9} 5^{3-9} x^{-1+3} y^{2+2} z^{4+1})^2$$
$$= (2^{11} 3^1 5^{-6} x^2 y^4 z^5)^2$$

Notes 0.2.4 and 0.2.6

$$= 2^{22} 3^2 5^{-12} x^4 y^8 z^{10}$$

Note 0.2.1

$$\boxed{= \frac{2^{22} 3^2 x^4 y^8 z^{10}}{5^{12}}}$$

2

Factor completely.

2.1

$$4x^3y^2 + 6x^2y^3$$

Note 0.3.1

$$\boxed{= 2x^2y^2(2x + 3y)}$$

2.2

$$a^2b^3c^4 - a^3b^4c^5 + 2a^2b^4c^4$$

Note 0.3.1

$$\boxed{= a^2b^3c^4(1 - abc + 2b)}$$

2.3

$$9a^2 - 1$$

Note 0.3.2

$$\boxed{= (3a - 1)(3a + 1)}$$

2.4

$$16x^2 - 25y^2$$

Note 0.3.2

$$\boxed{= (4x - 5y)(4x + 5y)}$$

2.5

$$25x^4y^8 - 64z^2$$

Note 0.3.2

$$\boxed{= (5x^2y^4 - 8z)(5x^2y^4 + 8z)}$$

2.6

$$16x^4 - 1$$

Note 0.3.2

$$= (4x^2 - 1)(4x^2 + 1)$$

Note 0.3.2

$$\boxed{= (2x - 1)(2x + 1)(4x^2 + 1)}$$

2.7

$$(s-2t)^2-4$$

Note 0.3.2

$$\boxed{=(s-2t-2)(s-2t+2)}$$

2.8

$$(x+3)^2 - (y+2)^2$$

Note 0.3.2

$$= (x+3-y-2)(x+3+y+2)$$

$$\boxed{= (x-y+1)(x+y+5)}$$

2.9

$$x^2(x^2-1)-9(x^2-1)$$

Note 0.3.1

$$=(x^2-9)(x^2-1)$$

Note 0.3.2

$$\boxed{=(x-3)(x+3)(x-1)(x+1)}$$

2.10

$$8x^9 + z^{12}$$

Note 0.3.4

$$\boxed{= (2x^3 + z^4)(4x^6 - 2x^3z^4 + z^8)}$$

2.11

$$8j^3 - 125k^6$$

Note 0.3.4

$$\boxed{= (2j - 5k^2)(4j^2 + 10jk^2 + 25k^4)}$$

2.12

$$(x-y)^3 - 1$$

Note 0.3.4

$$\boxed{= (x-y-1)((x-y)^2 + (x-y) + 1)}$$

2.13

$$x^2 - 2x - 3$$

Note 0.3.5

$$\boxed{= (x-3)(x+1)}$$

2.14

$$s^2 + 7s + 10$$

Note 0.3.5

$$\boxed{= (s+5)(s+2)}$$

2.15

$$48 - 13q - q^2$$

$$= -(q^2 + 13q - 48)$$

Note 0.3.5

$$\boxed{= -(q + 16)(q - 3)}$$

2.16

$$z^6 - 6z^3 - 72$$

Note 0.3.5

$$\boxed{= (z^3 - 12)(z^3 + 6)}$$

2.17

$$t^4 - 9st^2 - 10s^2$$

Note 0.3.5

$$\boxed{= (t^2 - 10s)(t^2 + s)}$$

2.18

$$a^6b^4 - a^3b^2c - 42c^2$$

Note 0.3.5

$$\boxed{= (a^3b^2 - 7c)(a^3b^2 + 6c)}$$

2.19

$$4x^2 - 4x + 1$$

Note 0.3.3

$$\boxed{= (2x-1)^2}$$

2.20

$$4n^2 - 12n + 9$$

Note 0.3.3

$$\boxed{= (2n-3)^2}$$

2.21

$$2r^2 - r - 3$$

Note 0.3.6

$$= 2r^2 - 3r + 2r - 3$$

$$\boxed{= (r+1)(2r-3)}$$

2.22

$$9a^2 + 15ab + 4b^2$$

Note 0.3.6

$$= 9a^2 + 12ab + 3ab + 4b^2$$

$$\boxed{= (3a + b)(3a + 4b)}$$

2.23

$$7s^6 - 9s^3 + 2$$

Note 0.3.6

$$\begin{aligned} &= 7s^6 - 7s^3 - 2s^3 + 2 \\ &= (7s^3 - 2)(s^3 - 1) \end{aligned}$$

Note 0.3.4

$$\boxed{= (7s^3 - 2)(s - 1)(s^2 + s + 1)}$$

2.24

$$a^2 + ab + 5a + 5b$$

Note 0.3.6

$$\boxed{= (a+5)(a+b)}$$

2.25

$$x^3 - x^2 - x + 1$$

Note 0.3.6

$$= (x^2 - 1)(x - 1)$$

Note 0.3.2

$$\boxed{= (x-1)^2(x+1)}$$

2.26

$$64x^4 + 1$$

$$= 64x^4 + 16x^2 + 1 - 16x^2$$

Note 0.3.3

$$= (8x^2 + 1)^2 - 16x^2$$

Note 0.3.2

$$\boxed{= (8x^2 - 4x + 1)(8x^2 + 4x + 1)}$$

2.27

$$4m^8 - 4m^4 + 9$$

$$= 4m^8 + 12m^4 + 9 - 16m^4$$

Note 0.3.3

$$= (2m^4 + 3)^2 - 16m^4$$

Note 0.3.2

$$\boxed{= (2m^4 - 4m + 3)(2m^4 + 4m + 3)}$$

3

Perform the indicated operations and simplify.

3.1

$$\frac{3a-3b}{c^2+3cd}\cdot\frac{2cd+6d^2}{ac-bc}$$

$$=$$

3.2

$$\frac{x^3 - 8}{x^2 - 6x + 8} \div \frac{3x^3 + 6x^2 + 12x}{x^2 - 8x + 16}$$

$$=$$

3.3

$$\frac{x+1}{x} - \frac{y+1}{y}$$

$$=$$

3.4

$$\frac{a}{bc} + \frac{b}{ac} - \frac{c}{ab}$$

$$=$$

3.5

$$\frac{2}{x^2} + \frac{3}{xy} - \frac{1}{y^2}$$

$$=$$

3.6

$$\frac{1}{x} - \frac{2}{x+y}$$

$$=$$

3.7

$$\frac{2}{x-y}+\frac{3}{y-x}$$

$$=$$

3.8

$$\frac{2}{(a+b)(a-b)} + \frac{1}{(a-b)^2}$$

$$=$$

3.9

$$\frac{x^2}{x(x-1)} + \frac{1}{x} - \frac{1}{x-1}$$

$$=$$

3.10

$$\frac{4}{4x^2-9} - \frac{3}{2x^2-x-3}$$

$$=$$

3.11

$$\frac{3}{2x-6} - \frac{9}{x^2-9}$$

$$=$$

3.12

$$\frac{x - \frac{9}{x}}{1 - \frac{3}{x}}$$

$$=$$

3.13

$$\frac{2 - \frac{7}{x} - \frac{4}{x^2}}{3 - \frac{13}{x} + \frac{4}{x^2}}$$

$$=$$

3.14

$$\frac{w-2+\dfrac{w-2}{w+2}}{w-\dfrac{3w+12}{w+2}}$$

$$=$$

3.15

$$\frac{a^2 - b^2}{a^3 - b^3}$$

$$=$$

3.16

$$\frac{x^3 - x^2y + xy^2 - y^3}{x^6 + y^6}$$

$$=$$

3.17

$$\left(\frac{x}{x^2-1}-\frac{3}{x+1}\right)\div\frac{2x^2-x-3}{x^3-1}$$

$$=$$

3.18

$$\frac{\left(\frac{x}{x+y}+\frac{y}{x-y}\right)\left(\frac{x^2-xy}{x^4-y^4}\right)}{\frac{x}{x^2+2xy+y^2}}$$

$$=$$

3.19

$$\frac{\dfrac{3}{p+q}+\dfrac{1}{p-2q}}{1+\dfrac{p-q}{p-2q}}$$

$$=$$

4

Perform the indicated operations and simplify.

4.1

$$\frac{b^{\frac{2}{5}}b^{\frac{1}{3}}}{b^{\frac{2}{15}}}$$

$$=$$

4.2

$$\frac{24c^{-\frac{1}{2}}d^{\frac{2}{3}}}{18c^{-\frac{1}{7}}d^{-\frac{3}{5}}}$$

$$=$$

4.3

$$\frac{\left(2^{\frac{2}{3}}x^{\frac{1}{2}}y^{\frac{1}{6}}z\right)^{6}}{4x^{\frac{2}{3}}y^{\frac{9}{2}}}$$

$$=$$

4.4

$$(u^{\frac{1}{3}} + (uv)^{\frac{1}{6}} + v^{\frac{1}{3}})(u^{\frac{1}{6}} - v^{\frac{1}{6}})$$

$$=$$

4.5

$$\left(\frac{r^{\frac{4}{3}}s^{\frac{1}{2}}t^{-2}}{rs^{\frac{2}{7}}t^{\frac{3}{5}}}\right)^{-10}$$

$$=$$

4.6

$$\sqrt[3]{-8^4}$$

$$=$$

4.7

$$\sqrt{384x^3y^6}$$

$$=$$

4.8

$$\sqrt[4]{9x^8}$$

$$=$$

4.9

$$\sqrt{\sqrt[3]{4a^4b^4}}$$

$$=$$

4.10

$$\sqrt{\frac{25c^5}{12a^3b}}$$

$$=$$

4.11

$$\frac{2x^2y}{\sqrt[3]{12x^2y^4}}$$

$$=$$

4.12

$$\sqrt{2}\sqrt[3]{2}$$

$$=$$

4.13

$$\frac{x^2-1}{\sqrt{x+1}}$$

$$=$$

4.14

$$\frac{\sqrt{a^2 - b^2}}{\sqrt{a^2 + b^2}}$$

$$=$$

4.15

$$\sqrt[4]{36} - \sqrt{54} + \sqrt{96}$$

$$=$$

4.16

$$\frac{2-\sqrt{3}}{3-\sqrt{2}}$$

$$=$$

4.17

$$\frac{\sqrt{6}-\sqrt{5}}{\sqrt{5}+\sqrt{6}}$$

$$=$$

4.18

$$\frac{x^2 - 2x + 1}{\sqrt{x} + 1}$$

$$=$$

4.19

$$\frac{6}{\sqrt{3-\sqrt{3}}}$$

$$=$$

4.20

$$\frac{1}{\sqrt[3]{4}+\sqrt[3]{-27}}$$

$$=$$

4.21

$$\frac{2\sqrt{5}}{\sqrt{8}} + \frac{9}{\sqrt[3]{16}}$$

$$=$$

5

Perform the indicated operations and simplify.

5.1

$$\overline{1-i}$$

$$=$$

5.2

$$\overline{5i+2}$$

$$=$$

5.3

$$(\overline{5-i}) - (\overline{5+i})$$

$$=$$

5.4

$$\sqrt{-1} - \sqrt{3} + \sqrt{2} - \sqrt{-4}$$

$$=$$

5.5

$$4 - 2\sqrt{-45} - (\overline{3 - \sqrt{-20}})$$

$$=$$

5.6

$$\sqrt{-12}\sqrt{-75}$$

$$=$$

5.7

$$3i(i^2 - i^3 + 5i^5 - i^{-2})$$

$$=$$

5.8

$$(3-5i)(7+4i)$$

$$=$$

5.9

$$(1+2i)(3-4i)$$

$$=$$

5.10

$$(i-1)(\overline{5+3i})$$

$$=$$

5.11

$$(2i-3)^2$$

$$=$$

5.12

$$\frac{3+4i}{2-i}$$

$$=$$

5.13

$$\frac{\sqrt{-9}+\sqrt{8}}{\sqrt{-1}}$$

$$=$$

5.14

$$\frac{3i-2}{3i+2}$$

$$=$$

5.15

$$\frac{(1-i)(3i+1)}{2i-1}$$

$$=$$

5.16

$$\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)\left(\frac{\sqrt{2}}{2}-\frac{\sqrt{2}}{2}i\right)$$

$$=$$

5.17

$$\frac{7+i-4(3-i)}{6-5i^3}$$

$$=$$

5.18

$$\frac{2-2(\overline{i+1})}{2-\sqrt{-4}}$$

$$=$$